HOANG KIM THIEN

Computer Science Student & Artificial Intelligence Researcher

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CAREER OBJECTIVE

Seeking research assistantships, graduate opportunities (Master's/PhD), and engineering roles in Artificial Intelligence, Computer Vision, and Machine Learning. Eager to contribute to efficient deep learning quantization, edge AI deployment, and text analytics while aiming to develop scalable and accessibility-focused AI systems for real-world applications.

EDUCATION

University of Sciences, Hue University — **2022 — 2026**

Bachelor of Science in Computer Science — Thua Thien Hue, Vietnam

- **Academic Standing:** Cumulative GPA of **3.83 / 4.00** (10-point scale: 9.08 / 10.00). Graduation Classification: **Excellent**.
- **Academic Ranking:** Ranked **2nd** among all undergraduate Information Technology students in the cohort (K46).
- **Scholarships & Honors:** Recipient of the prestigious University Merit-Based Scholarship in every semester for outstanding academic performance.

RESEARCH PUBLICATIONS

IEEE Conference Proceedings — **Accepted (Forthcoming 2026)**

The 9th International Conference on Multimedia Analysis and Pattern Recognition (MAPR) — Hue, Vietnam

- **Paper Title:** "Fast-Converging and Architecture-Agnostic 6-Bit Face Recognition via Gradient Coordination and Refined Data".
- **Authors:** Hoang Kim Thien and Le Quang Chien.
- **Key Contributions:** Developed an architecture-agnostic 6-bit (Q6) quantization framework for deep face recognition. Introduced a β-weighted Mean Squared Error (β-MSE) loss function to align quantized student gradients with a full-precision teacher, along with a stable three-phase training recipe.
- **Key Achievements:** Replaced massive synthetic training datasets with a compact subset of 53,500 real images (110x fewer). Recovered 102% of the teacher's baseline performance on the AgeDB-30 dataset while achieving **57.4x faster convergence** (3,135 vs. 180,000 iterations).

AWARDS & HONORS

- **Best Paper Award** — Faculty Student Scientific Research Conference, Hue University. (2026)
- **Second Prize** — national WCAG 2.2 Accessible Website Design Competition. (2025)
- **Second Prize** — University ICPC (International Collegiate Programming Contest) Contest. (2024)
- **Consolation Prize** — National Mathematical Olympiad for University Students in Linear Algebra. (2024)
- **University Merit Scholarships** — Awarded in consecutive semesters for top academic performance. (2022 — 2026)

PROJECTS & DEVELOPMENT EXPERIENCE

6-Bit Quantized Face Recognition **2025 — 2026**

Undergraduate Thesis Project & Research PyTorch, Computer Vision, Model Quantization

- Implemented low-bit quantization-aware training (QAT) to compress face embeddings to 6-bit representations, achieving a theoretical 5.33x storage size reduction for resource-constrained edge AI devices.
- Evaluated robust performance across multiple backbone networks including iResNet-18, iResNet-50, and MobileFaceNet on standard face verification benchmarks (LFW, CFP-FP, AgeDB-30, CALFW, CPLFW).

SmartAgri: Accessible Digital Farming Platform **2025**

Award-Winning Accessibility Design Submission Next.js, TailwindCSS, Firebase, OpenAI API, WCAG 2.2

- Designed a full-stack digital agriculture ecosystem structured for elderly, illiterate, and disabled farmers, implementing a strict "Accessibility-First" layout compliant with WCAG 2.2 AA.
- Integrated an AI agricultural advisor bot (via GPT API), a screen-reader compatible layout (Semantic HTML, ARIA regions), custom skip links, focus management, and a natural voice reading toolbar. Won 2nd Prize nationally.

Vietnamese Text Summarization Pipeline **2025**

Natural Language Processing Course Project Python, NLP, viT5, Transformers, TextRank

- Constructed a hybrid summarization engine combining extractive sentences filtering (TextRank with average Word2Vec embeddings) and abstractive transformer sequence generation (viT5-base).
- Conducted ablation studies analyzing the "lost-in-the-middle" effect on context lengths (512 vs 1024 tokens), proving that a 512-token input limit improves ROUGE-1 scores to 60.33 and ROUGE-L to 41.22 on 48,970 samples.

Containerized Hadoop Cluster & Network Routing **2024**

Distributed Computing Research Project Hadoop 3.2.1, Docker, Docker Compose, Linux Networking

- Deployed a distributed containerized HDFS cluster. Diagnosed and resolved NAT network issues preventing external hosts from reading/writing HDFS blocks due to DataNode dynamic container IP mapping.
- Implemented custom Docker bridge subnets, assigned static IP allocations (`172.22.0.2` and `172.22.0.3`), and mapped fixed hostnames in Docker files to allow external client connectivity.

Recyclable Waste Active Detection System **2025**

Computer Vision Course Term Paper YOLOv8, PyTorch, Active Learning, OpenCV

- Created a real-time object detection model to identify six classes of recyclable waste (cardboard, glass, metal, paper, plastic, biodegradable) for robotic sorting lines.
- Designed a pool-based active learning loop utilizing least-confidence query strategies, increasing mAP50-95 from 0.0616 (Round 0) to 0.2131 (Round 3) with only 1,332 annotated images, decreasing labeling cost.

TECHNICAL SKILLS

Languages	Python, C++, Java, JavaScript, TypeScript, SQL (MySQL, PostgreSQL, SQLite)
AI & Data Science	PyTorch, TensorFlow, Scikit-learn, Ultralytics YOLOv8, viT5, Hugging Face Transformers, Active Learning, OpenCV, Optuna, SHAP, LIME
Web Engineering	React.js, Next.js, Node.js, TailwindCSS, CSS Grid/Flexbox, Firebase, Vercel
Tools & Cloud	Docker, Docker Compose, Linux/Ubuntu, Git, GitHub, Google Colab, WSL, WebHDFS

LEADERSHIP & EXTRACURRICULARS

HUSC Chess Club & Social Work Team **2023 — Present**

Vice President (Chess Club) & Volunteer Hue University, Vietnam

- **Vice President of HUSC Chess Club:** Managed budget, scheduled matches, trained members, and co-hosted regional university championships. Awarded Certificate of Outstanding Leadership.
- **Social Work & Volunteering:** Devoted 50+ hours in charity campaigns and faculty volunteering events (e.g. IT Chef, IT Talent campaigns) supporting community health and student engagement.

LANGUAGES

- **Vietnamese:** Native (Mother tongue).
- **English:** Professional Working Proficiency. British Council Aptis ESOL Certificate - **Overall CEFR B2** (Listening: C1, Reading: B2, Speaking: B2, Writing: B1).
- **Japanese:** Elementary / Beginner (N4/N5 currently studying).

RESEARCH INTERESTS

Computer Vision, Model Compression & Quantization, Knowledge Distillation, Explainable AI (XAI), Natural Language Processing, Assistive & Accessibility Tech (WCAG), Reinforcement Learning & Wireless MAC Network Optimizations (RFID).